

Use short division with two-digit divisors

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $3,744 \div 12$. Copy or complete the first step.

2. Calculate $5,616 \div 18$.

3. Calculate $8,064 \div 21$.

4. Miro says: Miro uses a one-digit division fact and forgets the full two-digit divisor. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $3,744 \div 12$. Copy or complete the first step.

Answer 312.

2. Calculate $5,616 \div 18$.

Answer 312.

3. Calculate $8,064 \div 21$.

Answer 384.

4. Miro says: Miro uses a one-digit division fact and forgets the full two-digit divisor. Is this correct? Explain.

Answer Each quotient digit must represent a multiple of the complete divisor.

Use short division with two-digit divisors

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $3,744 \div 12$.

6. Check this result using an inverse, estimate or known fact:
Calculate $8,064 \div 21$.

2. Calculate $5,616 \div 18$.

7. Explain why short division is efficient for $9,600 \div 24$.

3. Calculate $8,064 \div 21$.

8. Identify and correct this misconception: Miro uses a one-digit division fact and forgets the full two-digit divisor.

4. Solve this independently and show every stage: Calculate $3,744 \div 12$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $5,616 \div 18$.

10. Write and solve a similar question to: Calculate $8,064 \div 21$.

Teacher answers and checking prompts

1. Calculate $3,744 \div 12$.
Answer 312.

2. Calculate $5,616 \div 18$.
Answer 312.

3. Calculate $8,064 \div 21$.
Answer 384.

4. Solve this independently and show every stage: Calculate $3,744 \div 12$.
Answer 312.

5. Use a second method or representation for: Calculate $5,616 \div 18$.
Answer Expected result: 312. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate $8,064 \div 21$.
Answer Expected result: 384. A valid check must be shown.

7. Explain why short division is efficient for $9,600 \div 24$.
Answer Known multiples of 24 make each stage manageable; the quotient is 400.

8. Identify and correct this misconception: Miro uses a one-digit division fact and forgets the full two-digit divisor.
Answer Each quotient digit must represent a multiple of the complete divisor.

9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate $8,064 \div 21$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use short division with two-digit divisors

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Explain why short division is efficient for $9,600 \div 24$.

6. Create a new example that tests the same idea as: Calculate $8,064 \div 21$.

2. Prove the answer to this question, rather than only calculating it: Calculate $3,744 \div 12$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $5,616 \div 18$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 384.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses a one-digit division fact and forgets the full two-digit divisor.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Explain why short division is efficient for $9,600 \div 24$.
Answer Known multiples of 24 make each stage manageable; the quotient is 400.
2. Prove the answer to this question, rather than only calculating it: Calculate $3,744 \div 12$.
Answer Expected result: 312. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $5,616 \div 18$.
Answer Expected result: 312. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 384.
Answer One valid reconstruction is: Calculate $8,064 \div 21$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses a one-digit division fact and forgets the full two-digit divisor.
Answer Each quotient digit must represent a multiple of the complete divisor.
6. Create a new example that tests the same idea as: Calculate $8,064 \div 21$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.